

Total No. of Questions :6]

SEAT No. :

P 67

[Total No. of Pages : 2

APR-17/BE/Insem-77

B.E. (Information Technology) (Semester - II)

REAL TIME & EMBEDDED SYSTEMS

(2012 - Pattern) (Elective - IV (b))

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidate:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks*
- 4) Assume suitable data if necessary*

UNIT - I

- Q1)** a) What is embedded system? Explain with the help of block diagram. [5]
- b) Discuss in brief the typical characteristics & design challenges of embedded system. [5]

OR

- Q2)** a) Discuss the advantages and disadvantages of top-down and bottom-up design process in embedded system design. [5]
- b) Define embedded system. List and discuss different types of embedded systems. [5]

UNIT - II

- Q3)** a) Explain in brief DMA Structure in SHARC processor and possible data operations in SHARC processor. [5]
- b) Explain the main features of ARM processor with core architectural block diagram. [5]

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OR

- Q4)** a) Explain exception handling process in ARM processors. [5]
- b) What are the key features of SHARC processor and discuss the targeted application areas for this processor. [5]

UNIT - III

- Q5)** a) Explain bus arbitration algorithm for ethernet bus. [5]
- b) What are main features of CAN2.0 bus standards explain along with specifications of the bus. [5]

OR

- Q6)** a) Explain the message protocol in I2C bus with respect to master and slave bidirectional communication. [5]
- b) Explain myrinet with suitable myrinet network diagram. Discuss the characteristics that distinguish myrinet from other networks [5]

